

Branch: MCA (Data Science) Kargil	Semester: 2
Student Name: Chaitanya Sharma	UID: 25MCD10056
Subject Name: Technical Training - I Lab	Subject Code: 25CAP-652
Section/Group: 25MCD KAR-1	Date of Performance: 21-04-2026

Experiment 10

Experiment Aim:

To implement Transaction Control Language (TCL) commands in PostgreSQL using COMMIT, ROLLBACK, and SAVEPOINT for maintaining data consistency and controlling transactions. **Tools used:** PostgreSQL **Objectives:**

- **Transaction Management:** To understand the use of TCL commands in PostgreSQL.
- **Rollback Operations:** To revert unwanted changes using ROLLBACK.
- **Savepoint Usage:** To create checkpoints within transactions using SAVEPOINT.
- **Data Integrity:** To maintain consistency during multiple database operations.

Theory:

Transaction Control Language (TCL) commands are used to manage transactions in a database. A transaction is a sequence of SQL operations performed as a single logical unit. PostgreSQL supports TCL commands such as COMMIT, ROLLBACK, and SAVEPOINT to ensure reliability, consistency, and recovery in case of errors.

Experiment Steps:

Step 1: Creating Table

Query:

```
CREATE TABLE Payroll ( emp_id INT  
PRIMARY KEY, emp_name VARCHAR(50),  
salary DECIMAL(10,2) CHECK(SALARY > 0)  
);
```

Step 2: Inserting Sample Data

Query:

```
INSERT INTO Payroll VALUES  
(1, 'Amit', 30000),  
(2, 'Neha', 40000),  
(3, 'Ravi', 50000);  
SELECT * FROM Payroll;
```

Output:

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Showing rows: 1 to 3 ✎ Page No: 1 of 1 ⏪			
	emp_id [PK] integer ✎	emp_name character varying (50) ✎	salary numeric (10,2) ✎
1	1	Amit	30000.00
2	2	Neha	40000.00
3	3	Ravi	50000.00

Step 3: Using ROLLBACK

Query: --

using rollback

begin;

UPDATE Payroll

SET salary = -1000

WHERE emp_id = 3;

ROLLBACK;

UPDATE Payroll

SET salary = 1000

WHERE emp_id = 3;

SELECT * FROM Payroll

Output:

Data Output Messages Notifications

ERROR: new row for relation "payroll" violates check constraint
"payroll_salary_check"
Failing row contains (3, Ravi, -1000.00).

SQL state: 23514

Detail: Failing row contains (3, Ravi, -1000.00).

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ROLLBACK

Query returned successfully in 131 msec.

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Showing rows: 1 to 3 ✎ Page No: 1 of 1 ⏪			
	emp_id [PK] integer ✎	emp_name character varying (50) ✎	salary numeric (10,2) ✎
1	1	Amit	30000.00
2	2	Neha	40000.00
3	3	Ravi	1000.00

Step 4: Using SAVEPOINT

Query:

BEGIN;

UPDATE Payroll

SET salary = salary + 5000

WHERE emp_id = 1;

SAVEPOINT sp1;

UPDATE Payroll

SET salary = salary + 7000

```
WHERE emp_id = 2;
```

```
UPDATE Payroll
```

```
SET salary = -1000
```

```
WHERE emp_id = 3;
```

```
ROLLBACK TO sp1;
```

```
SELECT * FROM Payroll;
```

```
COMMIT;
```

Output:

```
Data Output  Messages  Notifications
UPDATE 1
Query returned successfully in 529 msec.
```

```
Data Output  Messages  Notifications
SAVEPOINT
Query returned successfully in 112 msec.
```

Data Output Messages Notifications

ERROR: new row for relation "payroll" violates check constraint
"payroll_salary_check"

Failing row contains (3, Ravi, -1000.00).

SQL state: 23514

Detail: Failing row contains (3, Ravi, -1000.00).

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ROLLBACK

Query returned successfully in 81 msec.

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	emp_id [PK] integer 	emp_name character varying (50) 	salary numeric (10,2) 
1	2	Neha	40000.00
2	3	Ravi	1000.00
3	1	Amit	35000.00

Outcomes:

- I shall be able to use TCL commands in PostgreSQL.
- I shall understand transaction rollback mechanisms.
- I shall learn to use SAVEPOINT for partial rollback.
- I shall maintain data consistency during transactions.